LECTURE 13: MATH1061/7861
TOPICS: GCD, LCM, THE EUCLIDEAN ALGORITHM
DATE: TUESDAY MARCH 24, 2026

Greatest Common Divisor. Lowest Common Multiple.

Learning objectives:

(1) Understand the definitions of $\gcd(a, b)$ and $\mathrm{lcm}(a, b)$ for integers a and b.

(2) Apply the Euclidean algorithm to compute $\gcd(a, b)$.

Prime factorisation

$$n = p_1^{k_1} \cdots p_m^{k_m}$$

p_i's are prime.

$k_i > 0$.

Note: Euclidean Algorithm is much faster that prime factorisation.

Roughly speaking: Euclidean algorithm is linear whereas prime factorisation has exponential run-time.

Q1:

(a) Determine the prime factorisation of 63 and 105.

$$63 = 3 \times 21 = 3 \times 3 \times 7 = \underline{3^2 \times 7}$$

$$105 = 3 \times 35 = \underline{3 \times 5 \times 7}$$

(b) Use this to find the gcd and lcm of 63 and 105.

$$\gcd(63, 105) = 3 \times 7 = 21$$

$$\operatorname{lcm}(63, 105) = \frac{63 \times 105}{\gcd(63, 105)} = \frac{63 \times 105}{21}$$

$$= 3^2 \times 5 \times 7 = 315$$

$13 \times 17 \qquad 5 \times 17$

Q2: Use the Euclidean algorithm to find gcd(221, 85). $= 17$

$$221 = 85 \times 2 + 51$$

$$85 = 51 \times 1 + 34$$

$$51 = 34 \times 1 + \underline{17}$$

$$34 = 17 \times 2 + 0$$

(c) Use the Euclidean algorithm to find gcd(105, 63).

$$105 = 1 \times 63 + 42$$

remainder.

$$63 = 1 \times 42 + 21$$

$$42 = 2 \times 21 + 0$$

gcd(105,63).

What is the point of the Euclidean algorithm?

Hint: What is gcd(999999, 999997)? 1

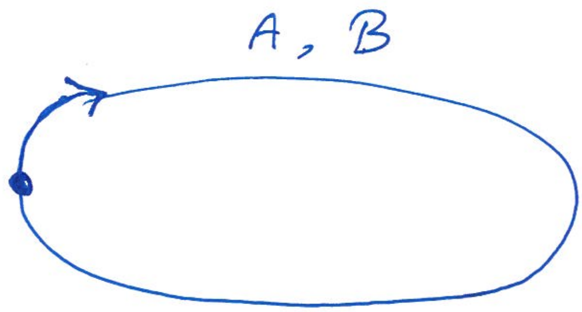
$$999999 = 1 \times 999997 + 2$$

$$999997 = 499998 \times 2 + 1$$

$$\cancel{499998 = 499998 \times 1 + 0}$$

$$2 = 2 \times 1 + 0$$

Q4: Two runners start running laps around a track. One completes a lap in 18 minutes, the other in 30 minutes. After how many minutes will they meet at the starting point again?



90 min

$90 = \text{lcm}(18, 30)$

A 18 min

B 30 min

A goes through ~~at~~ start point at multiples of 18: 0, 18, 36, 54, ...

B goes through start point at multiples of 30: 0, 30, 60, ...